

Pesticide & Pest Management

AgriMitra AI Knowledge Base — Document 3 of 6 — Integrated Pest Management Guide

Integrated Pest Management (IPM) Principles

- Monitor fields regularly (at least weekly) to catch pest build-up early, before economic damage occurs
- Use pest thresholds where available — treat only when pest population crosses a damaging level, not on a fixed calendar
- Prefer cultural and mechanical controls first: crop rotation, resistant varieties, pheromone traps, removal of infested plant parts
- Encourage natural enemies (ladybird beetles, spiders, parasitic wasps) by avoiding unnecessary broad-spectrum spraying
- Rotate chemical groups (not just brand names) across a season to slow resistance development
- Always follow the product label for dosage, timing, and pre-harvest interval — labels are the legal and safety authority, not general guides

Note for verification: This document intentionally avoids listing specific chemical names and per-litre dosages, since approved products, banned chemicals, and dosage recommendations change frequently and vary by state. Always verify current recommendations with your state agriculture department, ICAR, or a licensed pesticide dealer before advising a farmer on a specific product or dose.

Major Pests by Crop

Crop	Key Pests	Typical Damage Symptom
Paddy	Stem borer, Brown planthopper (BPH), White ear	Dead heart (stem borer); hopper burn patches (BPH); folded leaves (white ear)
Cotton	Pink bollworm, American bollworm, Whitefly, Jassids	Wilted, stunted flowers and bolls; sooty mold from whitefly honeydew; stippling (jassids)
Chilli	Thrips, Mites (mainly yellow mite), Red mite	Leaf curling and distortion (thrips/mites — 'murda' complex); holes (red mite)
Maize	Fall armyworm, Stem borer, Shoot fly	Window-pane feeding on leaves and whorl damage (fall armyworm); whorl damage (stem borer); shoot dieback (shoot fly)
Groundnut	Leaf miner, Spodoptera (armyworm), Thrips	Serpentine mines in leaves; defoliation (Spodoptera); silvering of leaf (thrips)
Sugarcane	Early shoot borer, Top borer, Pyrilla (leafhopper)	Dead heart (early shoot borer); bunched top (top borer); honeydew and sooty mold (pyrilla)

Broad Categories of Crop Protection Products

For orientation only — always match the specific product to the specific pest/disease using current, locally approved recommendations.

Category	Targets	General Note
Insecticides	Chewing and sucking insect pests	Choose selective products where possible to spare natural predators
Fungicides	Fungal diseases (blast, rust, blight, anthracnose)	Often most effective as a preventive spray before disease onset
Bactericides	Bacterial diseases (blight, wilt)	Copper-based products are commonly used; combine with resistant varieties
Herbicides	Weed control	Apply at the correct crop growth stage to avoid crop injury
Biopesticides	Pests and diseases, via natural agents (microbes, predators, parasitoids)	Lower risk, but often used within IPM programs alongside other methods

Safe Application Practices

- Always wear protective equipment: mask, gloves, long sleeves, and closed footwear while spraying
- Spray during early morning or late evening; avoid spraying in strong wind or before expected rain
- Never eat, drink, or smoke while handling or spraying pesticides
- Observe the pre-harvest interval (PHI) printed on the product label — do not harvest before this period has passed
- Triple-rinse empty containers and dispose of them safely; never reuse pesticide containers for food or water storage
- Wash hands, face, and exposed skin thoroughly with soap after spraying; wash sprayed clothing separately
- Store all pesticides in original, labeled containers, locked away and out of reach of children and livestock

Signs of Pesticide Poisoning (Farmer Safety)

If a person shows symptoms such as nausea, vomiting, dizziness, excessive sweating, blurred vision, or difficulty breathing after handling pesticides, treat this as a medical emergency: move them to fresh air, remove contaminated clothing, and seek immediate medical attention, taking the product label/container with them if possible.

Note for verification: This document is for general awareness. It does not replace the manufacturer's label, a licensed agronomist's advice, or your state's current list of approved/banned pesticides.